Potato Leaf Disease Classification: Model Performance Report

Dataset Overview

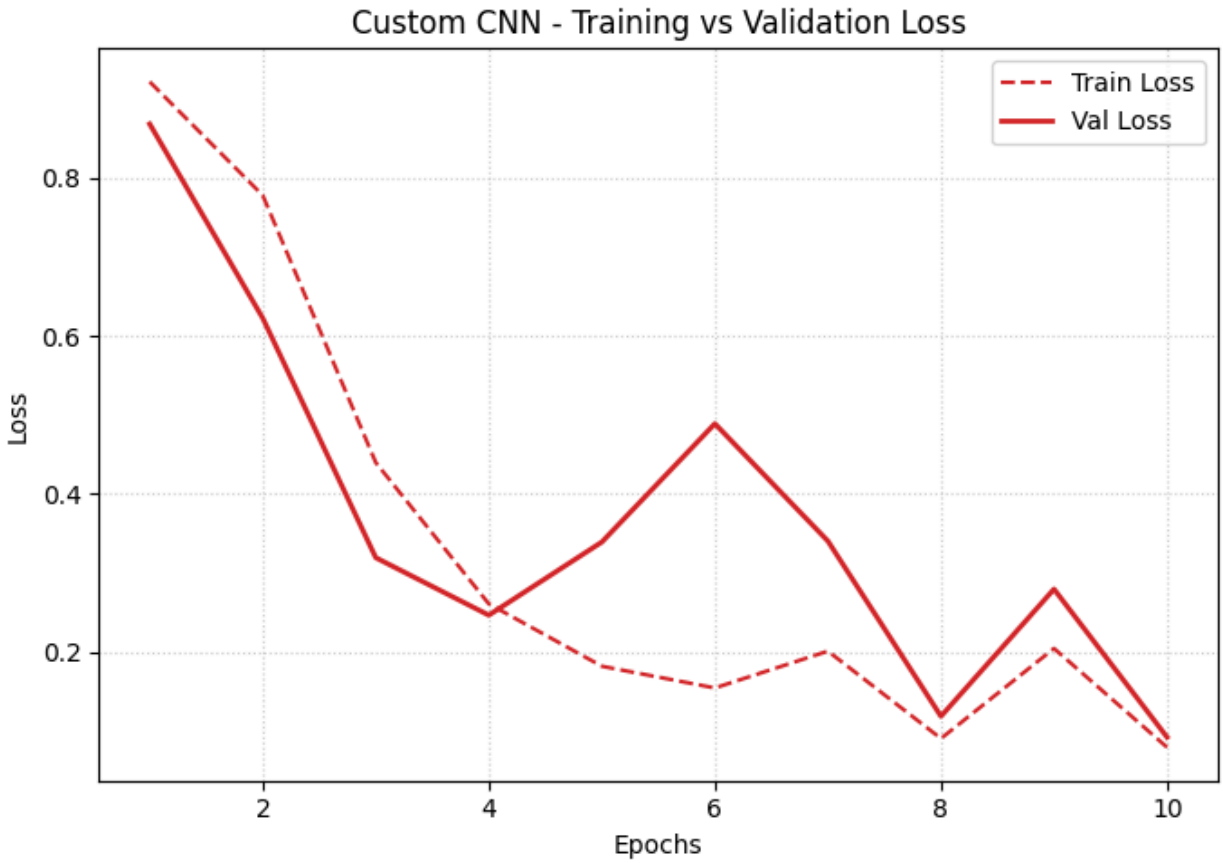
- **Classes:** 3 Potato Disease Classes (`Potato___Early_blight`, `Potato___Late_blight`, `Potato___healthy`)
- **Training Dataset:** ~1,728 images (80% split)
- **Validation Dataset:** ~430 images (20% split)
- **Training Strategy:** Single-stage transfer learning with frozen backbones for DenseNet121 and EfficientNetB3; end-to-end optimization from scratch for Custom CNN using `Adam` optimizer and `SparseCategoricalCrossentropy`.

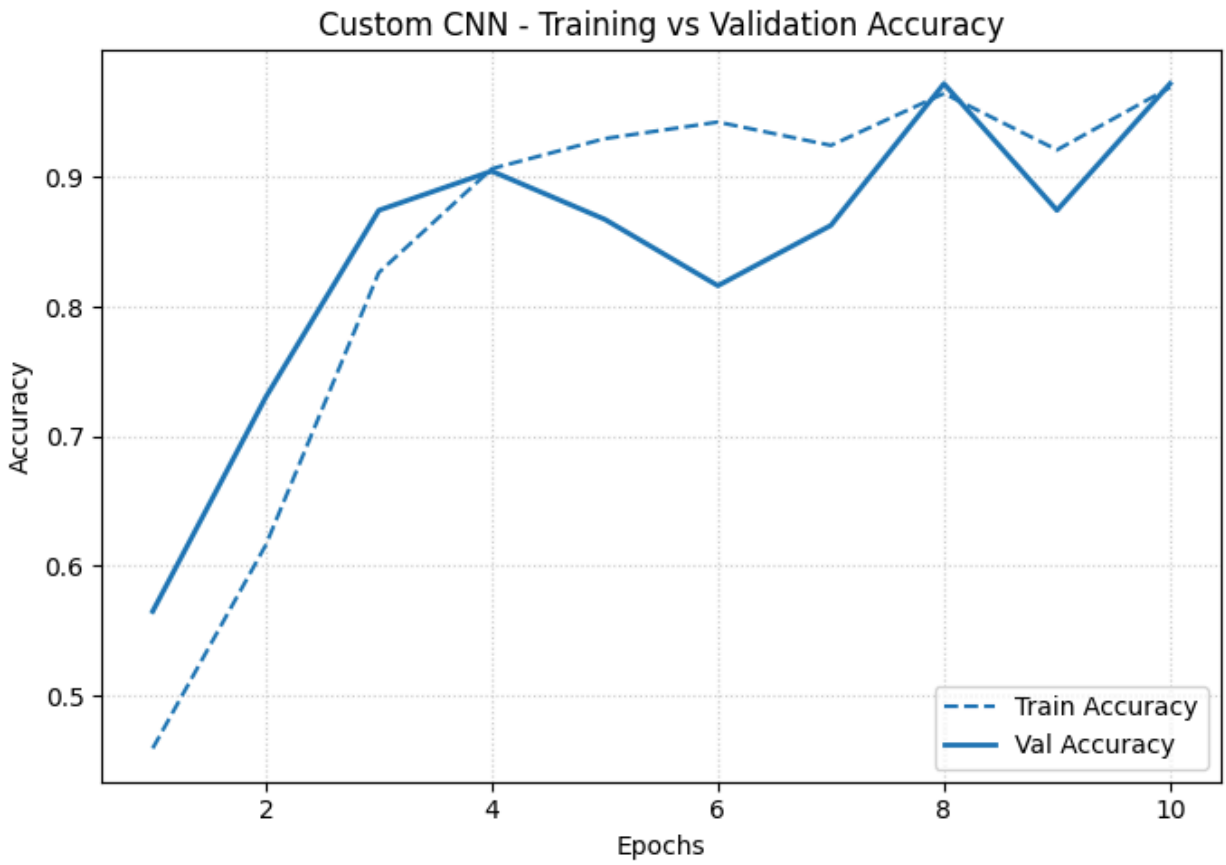
Model 1: Custom CNN Architecture

Metric / Parameter	Value
Input Resolution	224 *224 *3
Total Parameters	171,459 (0.17 M)
Trainable Parameters	171,459 (100%)
Peak Validation Accuracy	**97.21%**
Final Validation Loss	**0.0928**

Summary

The Custom CNN architecture provides a lightweight baseline using 6 repeated 3×3 convolutional and max-pooling blocks (171K parameters). Trained from scratch, it demonstrated rapid convergence, scaling from an initial 56.51% validation accuracy in Epoch 1 to a peak of 97.21% by Epoch 8 and 10. Mid-training exhibited minor validation loss spikes (up to 0.4890 in Epoch 6) before stabilizing at a final validation loss of 0.0928.





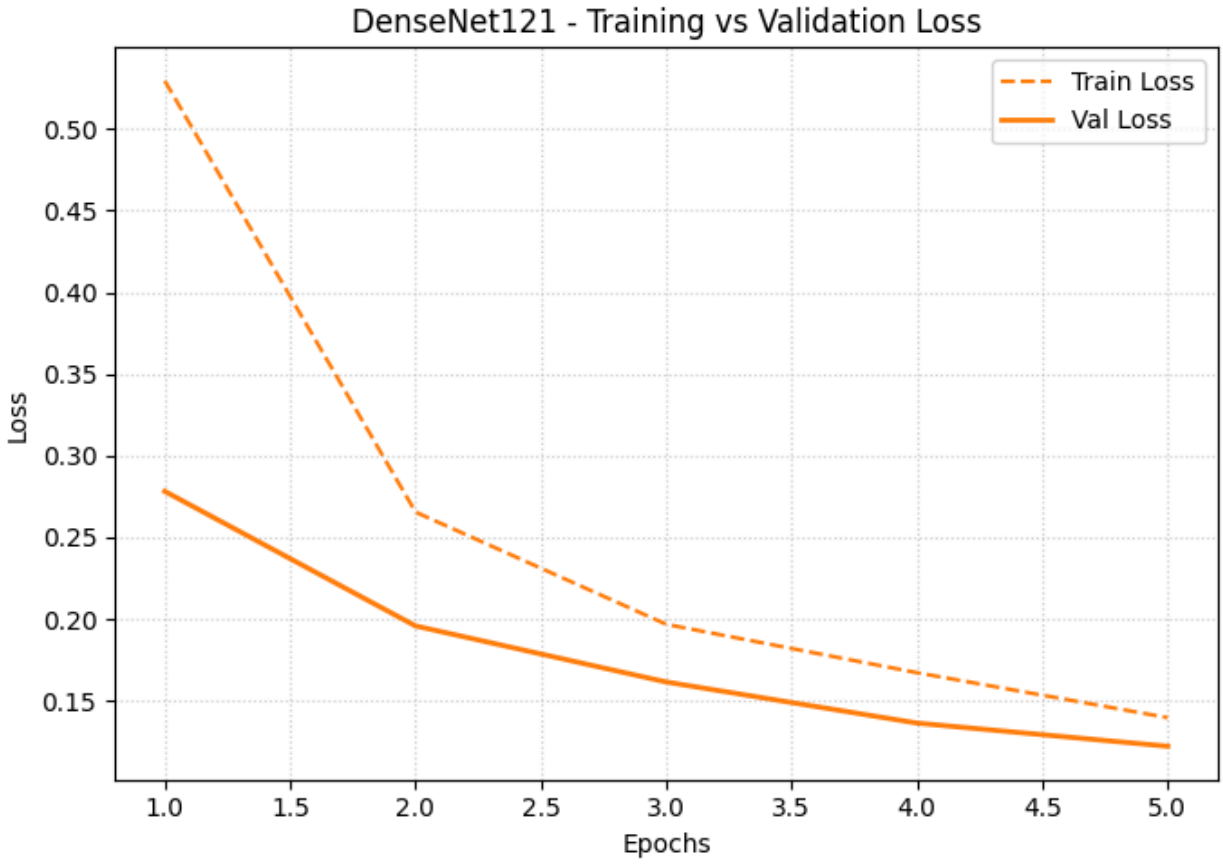
Model 2: DenseNet121

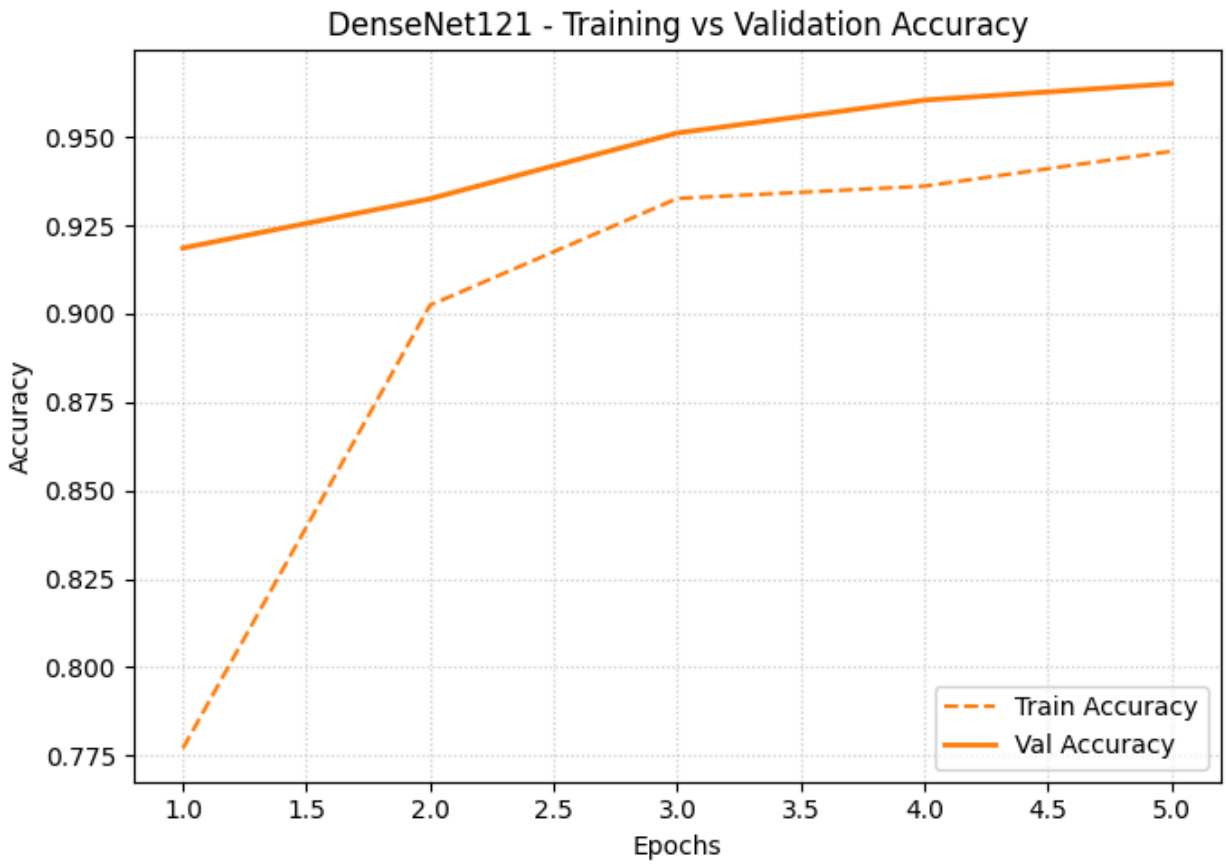
Metric / Parameter	Value
Input Resolution	224*224*3
Total Parameters	7,048,074 (7.05 M)
Trainable Parameters	~102,000 (Top Classification Head)
Peak Validation Accuracy	**96.51%**

Final Validation Loss	**0.1222**

Summary

DenseNet121 connects each layer to every subsequent layer in a feed-forward fashion, encouraging feature reuse across depth levels. Using pre-trained ImageNet weights with a frozen feature extraction backbone, the model achieved an initial validation accuracy of 91.86% in Epoch 1. Feature extraction showed steady epoch-over-epoch progress, pushing validation accuracy to 96.51% and lowering validation loss smoothly to 0.1222 by Epoch 5.





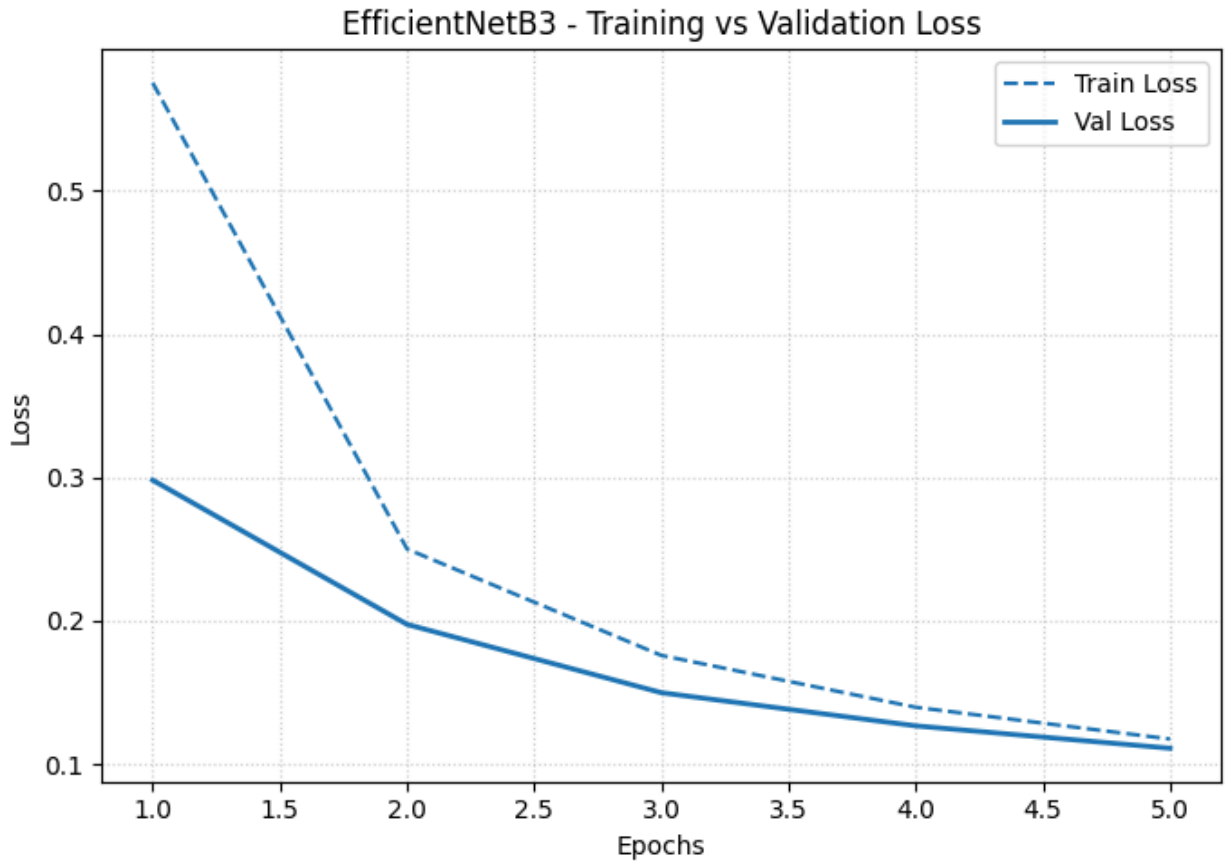
Model 3: EfficientNetB3

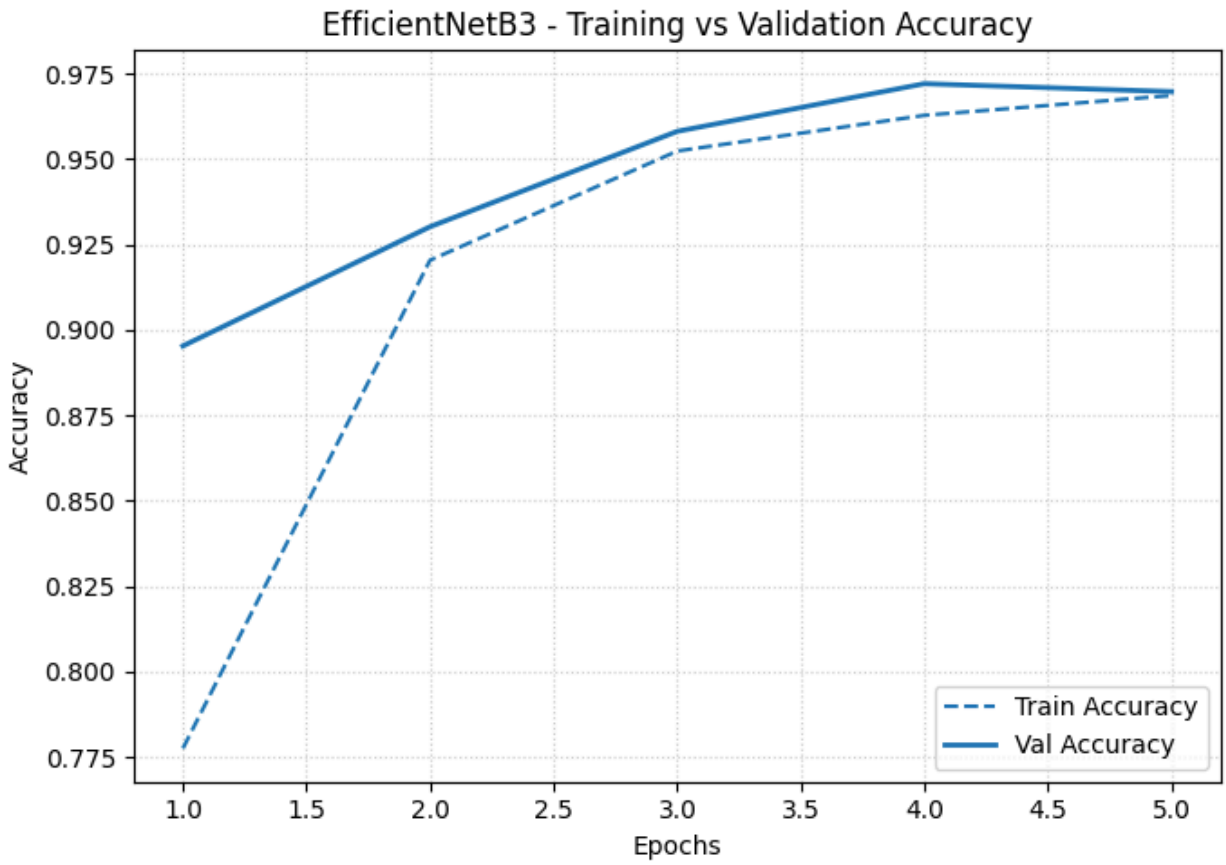
Metric / Parameter	Value
Input Resolution	224*224*3
Total Parameters	~12,238,000 (12.24 M)
Trainable Parameters	~115,000 (Top Classification Head)
Peak Validation Accuracy	**97.21%**

Final Validation Loss	**0.1110**

Summary

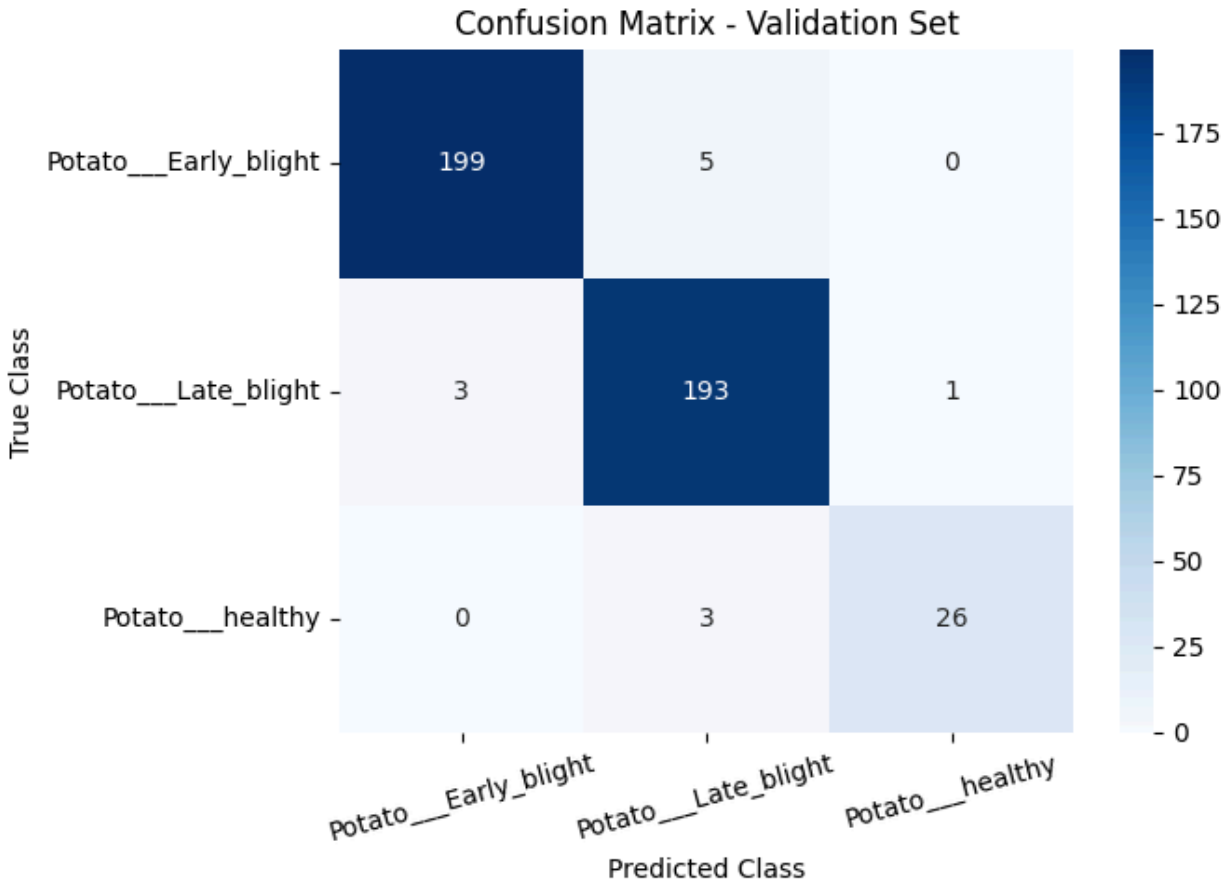
EfficientNetB3 utilizes compound scaling to systematically balance network depth, width, and resolution using Mobile Inverted Bottleneck Convolutions (MBConv). Operating with pre-trained weights, the model reached 89.53% validation accuracy in Epoch 1. During transfer learning feature extraction, performance scaled steadily across all 5 epochs, reaching a peak validation accuracy of 97.21% at Epoch 4 and a low validation loss of 0.1110.





Final Comparative Summary

Model	Parameters	Peak Val Accuracy	Final Val Loss
Custom CNN	0.17 M	97.21%	0.0928
DenseNet121	7.05 M	96.51%	0.1222
EfficientNetB3	12.24 M	97.21%	0.1110


Confusion Matrix - Validation Set
Potato___Early_blight
Potato___Late_blight
Potato___healthy
True Class
199
5
0
3
193
1
0
3
26
175
150
125
100
75
50
25
0
Potato___Early_blight
Potato___Late_blight
Potato___healthy
Predicted Class